Formalization of Formal Group Laws

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1 Formal Group Laws

1.1 Definition and basic properties of one dimensional Formal Group Laws

Definition 1.1.1. Let R be a commutative ring. A one dimensional Formal Group Law is a formal power series $F(X,Y) \in R[\![X,Y]\!]$ with the following properties:

1. $F(X,Y)$ is of the form
$$F(X,Y) = X + Y + \sum_{i,j\geq 1} c_{ij} X^i Y^j \tag{1.1.1.1}$$

2. $F(X,Y)$ satisfies associativity condition, namely
$$F(F(X,Y),Z) = F(X,F(Y,Z)) \tag{1.1.1.2}$$

Furthermore, if $F(X,Y)$ satisfies $F(X,Y) = F(Y,X)$, then it is called a commutative formal group law.

Remark 1.1.2. Under this definition, it can be proved that $F(X,0) = X$ and $F(0,Y) = 0$, which may be used as part of the definition in other references. Therefore, given a formal group law $F(X,Y) \in R[\![X,Y]\!]$, it can be written as $F(X,Y) = X + Y * G_1(X,Y)$ and $F(X,Y) = Y + X * G_2(X,Y)$, for some $G_1(X,Y), G_2(X,Y) \in R[\![X,Y]\!]$.

Proof of equivalence of the definition. Let $f(X) = F(X,0)$, then we have

$$F(F(X,0),0) = F(X,F(0,0)) = F(X,0)$$

which means that $f \circ f = f$. Composing both sides with f^{-1}, then we have $f = \mathrm{id}$. This is our desired result for $F(X,0) = X$. And $F(0,X) = 0$ holds via the similar proof. □

Example 1.1.3. There are two basic commutative formal group laws:

$$\mathbf{G}_a(X,Y) = X + Y \quad \text{(the additive formal group law.)} \tag{1.1.3.1}$$
$$\mathbf{G}_m(X,Y) = X + Y + XY \quad \text{(the multiplicative formal group laws.)} \tag{1.1.3.2}$$

Lemma 1.1.4. *Let $F(X,Y)$ be a formal group law over a ring R. Then there exists a power series $i(X) = -X + b_2X^2 + \cdots \in R[\![X]\!]$ such that $F(X,i(X)) = 0$ and $F(i(X),X) = 0$.*
In particular, the left inverse and right inverse is the same and the inverse is unique.

Proof. Let $i(X) = \sum_{j=1}^{\infty} b_j X^j$ and we inductively choose b_j to satisfy $F(X,i(X)) = 0$.

First, we choose b_1 to be -1 and then $F(X,i(X)) = 0 \bmod \deg 2$. Assume that $b_1, \ldots b_n$ is already defined and $F(X,i(X)) = 0 \mod \deg(n+1)$, then we choose b_{n+1} to be the minus $n+1$-th coefficient of $F(X,i_n(X))$, where $i_n(X) = \sum_{j=1}^{n} b_j X^j$. Therefore, $F(X,i_{n+1}(X)) = 0 \mod \deg(n+2)$, where $i_{n+1}(X) = i_n(X) + b_{n+1}X^{n+1}$. Repeating these process, we get the

desired series $i(X)$ and $F(X, i(X)) = 0$. We could use a similar process to define $i'(X)$ and $F(i'(X), X) = 0$.

Since $F(X, Y)$ is a formal group law, then we have

$$i'(X) = F(i'(X), 0) = F(i'(X), F(X, i(X))) = F(F(i'(X), X), i(X)) = F(0, i(X)) = i(X)$$

Therefore, 'left inverse' and 'right inverse' of X is the same. □

Combining the result above, given a formal group law $F(X, Y)$, then it defines a group structure for multivariate power series in $R[\![X_I]\!]$ with zero constant coefficient, where I is any finite index set.

Definition 1.1.5. Let $F(X, Y)$ and $G(X, Y)$ be two formal group laws over ring R. A *homomorphism* $F(X, Y) \to G(X, Y)$ is a power series $\alpha(X) = a_1X + a_2X^2 + \cdots \in R[\![X]\!]$ such that

$$\alpha(F(X, Y)) = G(\alpha(X), \alpha(Y)) \tag{1.1.5.1}$$

The homomorphism α is an *isomorphism* if there exists a homomorphism $\beta(X) : G(X, Y) \to F(X, Y)$ such that $\alpha(\beta(X)) = \beta(\alpha(X)) = X$.

Lemma 1.1.6. *The homomorphism $\alpha(X) : F(X, Y) \to G(X, Y)$ is an isomorphism if and only if a_1 is a unit in the ring, where $\alpha(X) = a_1X + a_2X^2 + \cdots$.*

Proof. Given that $\alpha(X)$ is a power series with a_1 unit, then we have a power series $\beta(X)$, such that $\alpha(\beta(X)) = \beta(\alpha(X)) = X$. And we should prove this $\beta(X)$ is a formal group homomorphism from $G(X, Y)$ to $F(X, Y)$, namely $\beta(G(X, Y)) = F(\beta(X), \beta(Y))$.

Since $\alpha(X)$ is homomorphism from $F(X, Y)$ to $G(X, Y)$, then

$$\alpha(F(\beta(X), \beta(Y))) = G(\alpha(\beta(X)), \alpha(\beta(Y))) = G(X, Y).$$

The desired equality is followed by pre-composing above equation with $\beta(X)$. □

Definition 1.1.7. Let $F(X, Y)$ be a formal group law over ring R and let $\phi : R \to R'$ be a ring homomorphism. Suppose $F(X, Y) = X + Y + \sum_{i,j\geq 1} c_{ij}X^iY^j$, then by applying ϕ to the coefficients of $F(X, Y)$ then we can obtain a formal group law

$$\phi_*F(X, Y) = X + Y + \sum \phi(c_{ij})X^iY^j \tag{1.1.7.1}$$

1.2 Functional Equation Lemma

In this section, we will discuss the functional equation lemma for formal group laws, and this can be used to construct some non-isomorphic formal group laws and construction of universal formal group law.

To begin with, we need some ingredients. Let R be a subring of K, $\sigma : K \to K$ is a ring homomorphism, I is an ideal of R, p is a prime number, q is a power of this prime number, and $s_1, s_2, \ldots$ are elements of K.

Those ingredients are suppose to satisfy the following conditions:

1. $\sigma(R) \subset R$

2. $\forall a \in R, \sigma(a) \equiv a^q \mod I$

3. $p \in I$

4. $\forall i,\ s_i I \subset R$

5. $\forall b \in K, r \in \mathbb{N},\ I^r b \subset I \Rightarrow I^r \sigma(b) \subset I$.

Definition 1.2.1. Let $g(X) = \sum_{i=1}^{\infty} b_i X^i$ be a power series in one variable with coefficients in R, then define the **recursion function** to be the unique solution of the following functional equation,

$$f_g(X) = g(X) + \sum_{i=1}^{\infty} s_i \sigma_*^i f_g(X^{q^i}) \tag{1.2.1.1}$$

where $\sigma_*^i f_g(X)$ is the power series obtained from $f_g(X)$ by applying the endomorphism σ^i to the coefficients.

Remark 1.2.2. Compared the coefficient both sides, there is a recursive formula for the coefficient of $f_g(X)$.

$$a_n = b_n + s_1 \sigma(a_{n/q}) + \cdots + s_r \sigma^r(a_{n/q^r}) = b_n + \sum_{i=1}^{r} s_i \sigma^i(a_{n/q^i}) \tag{1.2.2.1}$$

where $f_g(X) = \sum_{i=1}^{\infty} a_i X^i$ and $n = q^r m$, q does not divide m. In particular, if q does not divide n, then $a_n = b_n$.

Functional equation lemma is about the integrality of the coefficient of $f_g(X)$ and other power series related to this.

Before we start to discuss about the functional equation integrality lemma, let's discuss some technical lemma first.

Lemma 1.2.3. *Let $f_g(X) = \sum_{i=1}^{\infty} a_n X^n$, and let $n = q^r m$ where m is not divisible by q, then $a_n I^r \subset R$.*

Proof. By the condition for any given natural number r and $b \in K$,

$$I^r b \subset I \Rightarrow I^r \sigma(b) \subset I.$$

We have that for any given natural number j, r and $b \in K$,

$$I^r b \subset I \Rightarrow I^r \sigma^j(b) \subset I. \tag{1.2.3.1}$$

We do induction with respect to r (multiplicity of q in n).

Assume that the result holds for number less than $r+1$, now consider those n such that the multiplicity of q in n is $r+1$. According to our recursive formula, the coefficient a_n can be expressed as

$$a_n = b_n + \sum_{i=1}^{r+1} s_i \sigma^i(a_{n/q^i}).$$

Combining the result 1.2.3.1 and the property 4, it is enough to prove

$$\forall i \in [1, r+1], a_{n/q^i} I^{r+1} \in I.$$

According to induction hypothesis and the fact that the multiplicity of q in n/q^i is less than $r+1$, then $\forall i \in [1, r+1], a_{n/q^i} I^r \in R$. This implies that $a_{n/q^i} I^{r+1} \in I$. $\square$

Lemma 1.2.4. *Let $F(X_J), G(X_J) \in R[\![X_J]\!]$, where J is a finite index set, let r be the multiplicity of q in n, namely $n = q^r m$ with $q \nmid m$ and i be a positive natural number, assume that $F(X_J) \equiv G(X_J) \mod I^i$, then we have*

$$F(X_J)^n \equiv G(X_J)^n \mod I^{i+r}. \tag{1.2.4.1}$$

Proof. First we prove that $F(X_J)^{q^r} \equiv G(X_J)^{q^r} \mod I^{i+r}$, and then our goal is followed by taking pow at both sides.

The case for $r = 0$ is trivial. We do induction on r. Assume that the result hold for r, namely $F(X_J)^{q^r} \equiv G(X_J)^{q^r} \mod I^{i+r}$. Therefore, $F(X_J)^{q^r} = G(X_J)^{q^r} + H(X_J)$, where all coefficients of $H(X_J)$ are in I^{i+r}. And we have the following equality

$$\begin{aligned} F(X_J)^{q^{r+1}} = (F(X_J)^{q^r})^q &= (G(X_J)^{q^r} + H(X_J))^q \\ &= (G(X_J)^{q^r})^q + H(X_J)^q + qG(X_J)^{q^r} H(X_J)\alpha(X_J) \end{aligned}$$

for some $\alpha(X_J) \in R[\![X_J]\!]$. And the last equation follows from q is a prime power. Our result is given by the following two congruences $H(X_J)^q \equiv 0 \mod I^{i+r+1}$ and $qH(X_J) \equiv 0 \mod I^{i+r+1}$. □

Lemma 1.2.5. *For all $G(X_J) \in R[\![X_J]\!]$, where J is a finite index set, and let r be the multiplicity of q in n, namely $n = q^r m$ with $q \nmid m$, then*

$$G(X_J)^{nq^l} \equiv (\sigma_*^l G(X_J^{q^l}))^n \mod I^{r+1} \tag{1.2.5.1}$$

Proof. The case for $I = R$ is trivial. Now, we assume that $I \subsetneq R$. Then R/I is of characteristic p, since $p \in I$. Given that property 2, then we have

$$G(X_J)^{q^l} \equiv \sigma_*^l G(X_J^{q^l}) \mod I$$

We get the goal congruence by above congruence and lemma 1.2.4. □

Theorem 1.2.6. *Let $g(X) = \sum_{i=1}^{\infty} b_i X^i$, $h(X) = \sum_{n=1}^{\infty} c_n X^n$ be two power series in one variable over R, suppose that b_1 is invertible in R, then*

1. *the power series $F_g(X, Y) = f_g^{-1}(f_g(X) + f_g(Y))$ has its coefficients in R.*
2. *the power series $f_g^{-1}(f_h(X))$ has its coefficients in R.*
3. *there is a power series $\hat{h}(X) = \sum_{n=1}^{\infty} \hat{c}_n X^n$ with $\hat{c}_n \in R$ such that $f_g(h(X)) = f_{\hat{h}}(X)$.*
4. *if $\alpha(X) \in R[\![X]\!]$, $\beta(X) \in K[\![X]\!]$, and $r \in \mathbb{N}, r > 0$, then we have*

$$\begin{aligned} &\alpha(X) \equiv \beta(X) \mod I^r R[\![X]\!] \\ \Leftrightarrow\ &f_g(\alpha(X)) \equiv f_g(\beta(X)) \mod I^r R[\![X]\!] \end{aligned}$$

Remark 1.2.7. According to the functional equation lemma, $F_g(X, Y)$ is a commutative formal group law over coefficient ring R.

Proof of Functional Equation Lemma. Proof of part 1: Write $F(X,Y)$ and $f(X)$ for $F_g(X,Y)$ and $f_g(X)$. Let $F(X,Y) = F_1(X,Y)+F_2(X,Y)+\cdots$ where $F_i(X,Y)$ is homogeneous of degree i. Since $f(X) \equiv b_1 X \mod (\deg\ 2)$, then $f^{-1}(X) \equiv b_1^{-1} X \mod (\deg\ 2)$. Hence $F(X,Y) = X+Y \mod (\deg\ 2)$, i.e. $F_1(X,Y)$ has its coefficients in R. We now proceed by induction to prove that $F_n(X,Y) \in R[\![X,Y]\!]$.
Suppose that $F_1(X,Y),\ldots,F_{n-1}(X,Y) \in R[\![X,Y]\!]$. And we have

$$(F_1(X,Y)+\cdots F_{n-1}(X,Y))^r \equiv (F(X,Y))^r \mod (\deg\ (n+1)).$$

By the definition of $F(X,Y) = f^{-1}(f(X)+f(Y))$, we have

$$\begin{aligned} f(F(X,Y)) &= f(X)+f(Y) \\ \sigma_*^i f(\sigma_*^i F(X,Y)) &= \sigma_*^i f(X) + \sigma_*^i f(Y). \end{aligned}$$

by the recursion formula, $f(X) = g(X) + \sum_{n=1}^{\infty} s_n \sigma_*^n f(X^{q^n})$, substituting $F(X,Y)$ into X, and writing $f(X) = \sum_{n=1}^{\infty} a_n X^n$, we obtain that

$$f(F(X,Y)) = g(F(X,Y)) + \sum_{i=1}^{\infty} s_i \sum_{n=1}^{\infty} \sigma^i(a_n)(F(X,Y))^{q^i n} \tag{1.2.7.1}$$

Next by the second technical lemma, we have

$$s_i \sigma^i(a_n) F(X,Y)^{q^i n} \equiv s_i \sigma^i(a_n)(\sigma_*^i F(X^{q^i}, Y^{q^i}))^n \mod (R, \deg(n+1)) \tag{1.2.7.2}$$

then we have $\mod (R, \deg(n+1))$

$$\begin{aligned} f(F(X,Y)) &\equiv g(F(X,Y)) + \sum_{i=1}^{\infty} s_i \sum_{n=1}^{\infty} \sigma^i(a_n)(\sigma_*^i F(X^{q^i}, Y^{q^i}))^n \\ &\equiv g(F(X,Y)) + \sum_{i=1}^{\infty} s_i \sigma_*^i f(\sigma_*^i F(X^{q^i}, Y^{q^i})) \\ &\equiv g(F(X,Y)) + \sum_{i=1}^{\infty} s_i (\sigma_*^i f(X^{q^i}) + \sigma_*^i f(Y^{q^i})) \\ &\equiv g(F(X,Y)) + f(X) + f(Y) - g(X) - g(Y) \end{aligned}$$

Since $g(X) \equiv b_1 X \mod \deg\ 2$, then

$$g(F(X,Y)) \equiv b_1 F_n(X,Y) \mod (R, \deg(n+1))$$

use the equation $f(F(X,Y)) = f(X)+f(Y)$, and the congruence above, we have that

$$b_1 F_n(X,Y) \equiv 0 \mod (R, \deg(n+1))$$

and b_1 is invertible in R, then we get the result.

Proof of part 2: Write $G(X) = f_g^{-1}(f_{\bar{g}}(X))$ and $f(X) = f_g(X)$, then we have under $\mod (R)$

$$\begin{aligned}
f_{\bar{g}}(X) &= f(G(X)) \\
&= g(G(X)) + \sum_{i=1}^{\infty} s_i \sigma_*^i f(G(X)^{q^i}) \\
&= g(G(X)) + \sum_{i=1}^{\infty} s_i \sigma_*^i \sum_{i=1}^{\infty} a_n G(X)^{q^i n} \\
&\equiv g(G(X)) + \sum_{i=1}^{\infty} s_i \sum_{i=1}^{\infty} \sigma_*^i(a_n) \sigma_*^i G(X^{q^i})^n \\
&= g(G(X)) + \sum_{i=1}^{\infty} s_i \sigma_*^i f(G(X^{q^i})) \\
&= g(G(X)) + \sum_{i=1}^{\infty} s_i \sigma_*^i (f_{\bar{g}}(X^{q^i})) \\
&= g(G(X)) + f_{\bar{g}}(X) - \bar{g}(X)
\end{aligned}$$

where $f(X) = \sum a_i X^i$ And we can use the induction and the same argument in part 1 to prove all coefficient in $G(X)$ is in R.
Proof of part 3: Denote $\hat{f}(X) = f(h(X))$. By the functional equation 1.2.1.1, we only need to prove that $\hat{f}(X) - \sum_{i=1}^{\infty} s_i \sigma_*^i \hat{f}(X^{q^i})$ has all its coefficient in R. We have the following congruence:

$$\begin{aligned}
\hat{f}(X) - \sum_{i=1}^{\infty} s_i \sigma_*^i \hat{f}(X^{q^i}) &= f(h(X)) - \sum_{i=1}^{\infty} s_i \sigma_*^i f(\sigma_*^i h(X^{q^i})) \\
&= f(h(X)) - \sum_{i=1}^{\infty} s_i \sum_{n=1}^{\infty} \sigma^i(a_n)(\sigma_*^i h(X^{q^i}))^n \\
&\equiv f(h(X)) - \sum_{i=1}^{\infty} s_i \sum_{n=1}^{\infty} \sigma^i(a_n)(h(X)^{q^i n}) \\
&= f(h(X)) - \sum_{i=1}^{\infty} s_i \sigma_*^i f(h(X)^{q^i}) \\
&= g(h(X)) \equiv 0
\end{aligned}$$

all the congruence above is under the $\mod R$ sense. And the last term have all coefficient in R.

Proof of part 4: The implication direction : assume $\alpha(X) \equiv \beta(X) \mod I^r R[\![X]\!]$, then $\beta(X) \in R[\![X]\!]$. According to the lemma 1.2.4, we have that $\alpha(X)^n \equiv \beta(X)^n \mod I^{r+t}$, where t is the multiplicity of q in n. Let $f(X) = \sum_{i=1}^{\infty} a_i X^i$, according to the lemma 1.2.3, we have that $a_n \alpha(X)^t \equiv a_n \beta(X)^t \mod I^r$, where t is multiplicity of q in n. Therefore, $f(\alpha(X)) \equiv f(\beta(X)) \mod I^r$.
For the reverse direction, assume that $f(\alpha(X)) \equiv f(\beta(X)) \mod I^r R[\![X]\!]$.
First, we prove the following result,

$$\alpha(X) \equiv 0 \mod I^r \Rightarrow f^{-1}(\alpha(X)) \equiv 0 \mod I^r \tag{1.2.7.3}$$

Let $\gamma(X) = f^{-1}(\alpha(X))$, then $\alpha(X) = f(\gamma(X))$. Let $\gamma_n(X)$ be the truncation of degree $n+1$ of $\gamma(X)$, and we use induction to prove all $\gamma_n(X) \in I^r R[\![X]\!]$. For $k = 1, 2$ is trivial. Assume that

the result holds for k, namely $\gamma_k(X) \equiv 0 \mod I^r$ now we consider $\gamma_{k+1}(X)$. According to the functional equation and definition of $\gamma(X)$, we have

$$\alpha(X) = g(\gamma(X)) + \sum_{i=1}^{\infty} s_i \sigma_*^i f(\gamma(X)^{q^i}) \tag{1.2.7.4}$$

I would claim:

$$\sum_{i=1}^{\infty} s_i \sigma_*^i f(\gamma(X)^{q^i}) \in I^r R[\![X]\!] \mod \deg(k+2). \tag{1.2.7.5}$$

And we consider the $k+1$-th coefficient at both equation 1.2.7.4, then we have $b_1 * c_{k+1} \in I^r$, where $\gamma(X) = \sum_{i=1}^{\infty} c_i X^i$ and $g(X) = \sum_{i=1}^{\infty} b_i X^i$. Since b_1 is a unit in R, then $c_{k+1} \in I^r$.
Proof of claim 1.2.7.5 above. According to the lemma 1.2.4 and hypothesis assumption, we have that $\gamma_k(X)^n \equiv 0 \mod I^{r+t}$, where t is multiplicity of q in n. By the property 4, it is enough to prove that

$$\sigma_*^i f(\gamma(X)^{q^i}) \in I^{r+1} R[\![X]\!] \mod \deg(k+2) \tag{1.2.7.6}$$

which is equivalent to

$$\sigma^i(a_n)\gamma(X)^{q^i n} \in I^{r+1} R[\![X]\!] \mod \deg(k+2) \tag{1.2.7.7}$$

where $f(X) = \sum_{i=1}^{\infty} a_n X^n$. Since $\gamma(X)^{q^i n} \equiv \gamma_k(X)^{q^i n} \mod \deg(k+2)$ and lemma 1.2.3, we have the congruence 1.2.7.7. Now we proved the result 1.2.7.3.
By assumption $f(\alpha(X)) - f(\beta(X)) \in I^r R[\![X]\!]$, we have that $\delta(X) := f^{-1}(f(\beta(X)) - f(\alpha(X))) \in I^r R[\![X]\!]$. Then

$$\begin{aligned} f(\beta(X)) &= f(\delta(X)) + f(\alpha(X)) \\ \beta(X) &= f^{-1}(f(\delta(X)) + f(\alpha(X))) \end{aligned}$$

It is easy to prove that $f^{-1}(f(X) + f(Y))$ is a formal group law, then

$$\beta(X) = f^{-1}(f(\delta(X)) + f(\alpha(X))) = \alpha(X) + \delta(X)G(\delta(X), \alpha(X)). \tag{1.2.7.8}$$

for some $G(X,Y) \in R[\![X,Y]\!]$. Since $\delta(X) \in I^r R[\![X]\!]$, then $\beta(X) \equiv \alpha(X) \mod I^r R[\![X]\!]$.

□

1.3 Universal formal group laws

In this section, we will discuss the construction of universal formal group law and the applications of universal formal group laws.

Definition 1.3.1. A one dimensional formal group law $F(X,Y)$ over L is called a universal formal group law iff for any formal group law $G(X,Y)$ over R, there is a unique ring homomorphism $\phi : L \to R$, such that $\phi_* F(X,Y) = G(X,Y)$.

Remark 1.3.2. The ring L is uniquely determined up to isomorphism. Indeed, let $F(X,Y)$ and $F'(X,Y)$ is universal formal group law over ring L and L' respectively. Then there is a ring homomorphism $\phi : L \to L'$ and $\psi : L' \to L$ such that $\phi_* F(X,Y) = F'(X,Y)$ and $\psi_* F'(X,Y) = F(X,Y)$, then by uniqueness, we can deduce that $\phi\psi = \mathrm{id}_{L'}$ and $\psi\phi = \mathrm{id}_L$.

We can show that there exists a universal formal group law over some ring L. Let $\tilde{L} = \mathbb{Z}[C_{i,j}]$, where $C_{i,j}$ is intermediate and $i, j = 1, 2, \ldots$. Consider the power series $F_C(X, Y) = X + Y + \sum_{i,j} C_{i,j} X^i Y^j$. And we write

$$F_C(F_C(X, Y), Z) - F_C(X, F_C(Y, Z)) = \sum_{i,j,k} P_{i,j,k}(C) X^i Y^j Z^k$$

where $P_{i,j,k}(C)$ is some polynomial with variables $C_{i,j}$. Let I be the ideal of $\tilde{L}$ generated by $C_{i,j} - C_{j,i}$ and $P_{i,j,k}(C)$. Let $L = \tilde{L}/I$ and $F(X, Y) = p_* F_C(X, Y)$, where $p : \tilde{L} \to L$ be natural quotient map. Therefore, given a formal group law $F(X, Y)$ over ring R, if we denote $F(X, Y) = a_{ij} X^i Y^j$, then there is a unique ring homomorphism from $\phi : L \to R$ such that $\phi(C_{ij}) = a_{ij}$ and $\phi_* F_C(X, Y) = F(X, Y)$. The uniqueness is given by the equation $\phi_* F_C(X, Y) = F(X, Y)$.

However, we know nothing about the structure of ring L and $F(X, Y)$ above. In fact, we could prove that L is isomorphic to $\mathbb{Z}[U]$ and $F(X, Y) \in \mathbb{Z}[U][\![X]\!]$ is of the form $f^{-1}(f(X) + f(Y))$ for some $f(X) \in \mathbb{Q}[U][\![X]\!]$. And the integrality of the coefficients of $F(X, Y)$ follows by the functional equation lemma.

Remark 1.3.3. There is a functor from commutative ring category to type, which map commutative ring R to formal group over R and ring homomorphism $\phi : R \to S$ to pushforward map from formal groups over R to formal groups over S which is defined in definition 1.1.7. This is a covariant functor and is corepresented by ring L and $F_C(X, Y)$. To be specific, there is a bijection between $\mathcal{H}om_{Ring}(L, R)$ and formal group laws over R.

1.4 Commutativity theorem

In this section, we are going to prove that 'almost all' one dimensional formal group is commutative. To be more precisely,

Theorem 1.4.1. *Let R be a ring, then every one dimensional formal group law over ring R is commutative if and only if R has no element $a \neq 0$ that are nilpotent and $\mathbb{Z}$-torsion simultaneously.*

Before we start to prove the theorem, we should discuss some lemmas first.

Lemma 1.4.2. *Let $F(X, Y)$ be a non-commutative one dimensional formal group law over a ring R, then there is a nonzero homomorphism of $F(X, Y)$ into $\mathbf{G}_m(X, Y)$ or into $\mathbf{G}_m(X, Y)$.*

Proof. For simplicity, we denote $F(X, Y)$ as $X +_F Y$ and the inverse $i(X)$ as $-_F X$. Let $H(X, Y) = X +_F Y + (-_F X)$, we have $H(0, Y) = Y$. And since $F(X, Y)$ is non-commutative, then

$$H(X, Y) = Y + \sum_{i=1}^{\infty} r_n(X) Y^n \tag{1.4.2.1}$$

with $r_n(X) \neq 0$ for some n. Since $H(0, Y) = Y$, we have $r_n(0) = 0$ for all n. Due to the associativity of $F(X, Y)$, we have

$$H(X_1, H(X_2, Y)) = H(X_1 +_F X_2, Y) \tag{1.4.2.2}$$

Let i be the least integer such that $r_i(X) \neq 0$.
For the case $i = 1$, then

$$H(X, Y) \equiv (1 + r_1(X))Y \pmod{Y^2}$$

And combine with the equation 1.4.2.2, we have

$$H(X_1, H(X_2, Y)) \equiv (1 + r_1(X_1))H(X_2, Y) \equiv (1 + r_1(X_1))(1 + r_1(X_2))Y \pmod{Y^2} \quad (1.4.2.3)$$

$$H(X_1 +_F X_2, Y) \equiv (1 + r_1(X_1 +_F X_2))Y \pmod{Y^2} \quad (1.4.2.4)$$

Therefore, $(1 + r_1(X_1))(1 + r_1(X_2)) = 1 + r_1(X_1 +_F X_2)$, i.e.

$$r_1(X_1 +_F X_2) = r_1(X_1) + r_1(X_2) + r_1(X_1)r_1(X_2). \quad (1.4.2.5)$$

This means $r_1(X)$ is a formal group homomorphism from F to $\mathbf{G}_m$.

For the case $i > 1$, then we have

$$H(X, Y) \equiv Y + r_i(X)Y^i \pmod{Y^{i+1}}$$

According to the equation 1.4.2.2, we have

$$H(X_1, H(X_2, Y)) \equiv H(X_2, Y) + r_i(X_1)H(X_2, Y)^i \equiv Y + r_i(X_2)Y^i + r_i(X_1)Y^i \pmod{Y^i}. \quad (1.4.2.6)$$

$$H(X_1 +_F X_2, Y) \equiv Y + r_i(X_1 +_F X_2)Y^i \pmod{Y^{i+1}}. \quad (1.4.2.7)$$

Therefore, $r_i(X_1 +_F X_2) = r_i(X_1) + r_i(X_2)$, which means $r_i(X)$ is a formal group homomorphism from F to $\mathbf{G}_a$. □

Lemma 1.4.3. *Let R be a integral domain and let $F(X, Y), F'(X, Y)$ be one dimensional formal group laws over ring R. Suppose there exist a homomorphism $\alpha(X) : F(X, Y) \to F'(X, Y)$ and $F'(X, Y)$ is commutative, then $F(X, Y)$ is commutative.*

Proof. Let r be the order of $\alpha(X) = \sum_{n=1}^{\infty} a_n X^n$, i.e. the least n such that a_n does not equal to zero. Let $C(X, Y) = X +_F Y +_F (-_F X) +_F (-_F Y)$ be the commutator of X and Y with respect to F.

We can prove that a formal group law F is commutative if and only if the commutator $C(X, Y)$ with respect to F is zero.

Since F' is commutative, then

$$\alpha(C(X, Y)) = 0.$$

Suppose F is non-commutative, i.e. $C(X, Y) \neq 0$. Let $C_n(X, Y)$ is the homogeneous component in $C(X, Y)$ with degree n, and m be the least integer such that $C_m(X, Y)$ is non-zero. We have

$$\alpha(C(X, Y)) \equiv a_r C_m(X, Y)^r \pmod{\deg(mr + 1)}$$

Since $a_r \in R$ and R is a integral domain, then this term is non-zero, which contradicts to the assumption of $\alpha(C(X, Y)) = 0$. □

Proof of Commutative theorem 1.4.1. First, we prove the necessity of the condition on R.

Let $0 \neq a \in R$ be an element in R is a nilpotent and $\mathbb{Z}$-torsion, then there is formal group law $F(X, Y)$ over R such that $F(X, Y)$ is non-commutative.

Let n be the least integer such that $na = 0$, then $n > 1$ and there exists a prime number p, such that $p \mid n$. Let $b = \frac{n}{p}a$, then $b \neq 0$ and $pb = 0$, since n is minimal. Let m be the least integer such that $b^m = 0$ and $c = b^{m-1}$. Therefore, this c satisfies $c \neq 0, c^2 = 0, pc = 0$. Denote $F(X, Y) = X + Y + cXY^p$, then

$$F(F(X, Y), Z) = X + Y + Z + c(XY^p + XZ^p + YZ^p) = F(X, F(Y, Z))$$

Therefore, $F(X,Y)$ is a non-commutative formal group law over R.

Secondly, we prove sufficiency for the case R is an integral domain. Let $F(X,Y)$ be a formal group law over R, where R is an integral domain. Suppose that $F(X,Y)$ is non-commutative, then by lemma 1.4.2, there is a non zero formal group homomorphism from $F(X,Y)$ to $\mathbf{G}_a(X,Y)$ or $\mathbf{G}_m(X,Y)$. According to lemma 1.4.3 and $\mathbf{G}_a(X,Y), \mathbf{G}_m(X,Y)$ is commutative formal group law, then $F(X,Y)$ is commutative. This leads to a contradiction.

Thirdly, we prove the sufficiency for the case R is $\mathbb{Z}$-torsion free.

□

1.5 Height

In this section, we will discuss the definition and properties of height of a formal group law. Let $F(X,Y)$ be a commutative formal group law over ring R, fixed throughout this section.

Definition 1.5.1. For any for natural number n, we define n-series $[n](X) \in R[\![X]\!]$ as follows,

1. $[0](X) = 0$.
2. $[n+1](X) = F([n](X), X)$.

Proposition 1.5.2. *For any natural number n, n-series $[n](X)$ is a formal group homomorphism from $F(X,Y)$ to $F(X,Y)$, i.e. $[n](F(X,Y)) = F([n](X), [n](Y))$.*

Proof. We use indution on n to prove it. For the base case $n = 0, 1$, it follows easily. Assume that the result holds for some natural number k, now consider the case for $k+1$.

$$\begin{aligned}
[k+1]F(X,Y) &= F([k]F(X,Y), F(X,Y)) \\
&= F(F([k](X), [k]Y), F(Y,X)) \\
&= F([k](X), F([k](Y), F(Y,X)))
\end{aligned}$$

And then we focus on the term $F([k](Y), F(Y,X))$.

$$F([k](Y), F(Y,X)) = F(F([k](Y), Y), X) = F([k+1](Y), X) = F(X, [k+1](Y))$$

Then we have that

$$\begin{aligned}
[k+1]F(X,Y) &= F([k](X), F(X, [k+1](Y))) \\
&= F(F([k](X), X), [k+1](Y)) \\
&= F([k+1](X), [k+1](Y))
\end{aligned}$$

□

Remark 1.5.3. Via induction, we could deduce that $[n](X) = nX + O(X^2)$. As a result, for a ring R of characteristic p, we have $[p](X) = aX^k + O(X^{k+1})$ for some $k > 1$. Since $[p](X)$ is formal group endomorphism, then we have the following proposition

Proposition 1.5.4. *Assume R of characteristic p, then $[p](X) = 0$ or $[p](X) = aX^{p^n} + O(X^{p^n+1})$.*

From now on, we fix a prime number p,

Definition 1.5.5. Let v_n be the p^n-th coefficient of the p-series $[p](X)$. We say $F(X,Y)$ has height $\geq n$, if $v_i = 0$ for all $i < n$. And we say $F(X,Y)$ has height n if v_n is invertible in R. In particular, we say $F(X,Y)$ has infinity height if $[p](X) = 0$.

Remark 1.5.6. From the remark above, we have that $v_0 = p$. Then $F(X,Y)$ has height ≥ 1 if and only if $p = 0$ in R, which is equivalent to either R of characteristic p or R is trivial. And $F(X,Y)$ has height 0 if and only if p is invertible in R.

Example 1.5.7. Assume that R of characteristic p, the p-series for mulplicative formal group $\mathbb{G}_m(X,Y) = (1+X)(1+Y) - 1$ is $[p](X) = (1+X)^p - 1 = X^p$, then $\mathbb{G}_m(X,Y)$ has height 1. For the additive formal group law $\mathbb{G}_a(X,Y)$ the p-series $[p](X) = pX = 0$, then $\mathbb{G}_a(X,Y)$ has infinity height.

Proposition 1.5.8. *Let $F(X,Y)$ and $G(X,Y)$ be formal group laws over ring R, then $F(X,Y)$ and*

2 Lubin-Tate Theory

2.1 Definition and properties of Lubin-Tate formal group

Let K be a field complete with respect to a discrete valuation, with finite residue field. For instance, K is a nonarchimedean local field. Let $\mathcal{O}_K$ be the ring of integral of field K and let q be the cardinality of the residue field, π be a uniformizer of $\mathcal{O}_K$.

Definition 2.1.1. Let $\mathcal{F}_\pi$ denote the set of power series $f(X) \in \mathcal{O}_K[\![X]\!]$ which satisfy the two conditions:

$$f(X) \equiv \pi X \pmod{\deg 2} \tag{2.1.1.1}$$

$$f(X) \equiv X^q \pmod{\pi} \tag{2.1.1.2}$$

Lemma 2.1.2. *Let $f(X)$ and $g(X)$ be elements of $\mathcal{F}_\pi$ and let $L(X_1, \ldots, X_n) = \sum_{i=1}^n a_i X_i$ be a linear form with coefficients in $\mathcal{O}_K$. Then there exists $F(X_1, \ldots, X_n)$ with coefficients in $\mathcal{O}_K$ such that*

$$F(X_1, \ldots, X_n) \equiv L(X_1, \ldots, X_n) (\bmod \deg 2) \tag{2.1.2.1}$$

$$f(F(X_1, \ldots, X_n)) = F(g(X_1), \ldots, g(X_n)) \tag{2.1.2.2}$$

Proof. We use induction to prove that for all $k \in \mathbb{N}$, there exists a unique multivariate polynomial $F_k(X_1, \ldots, X_n)$ of degree k and

$$F_k(X_1, \ldots, X_n) \equiv L(X_1, \ldots, X_n) (\bmod \deg 2)$$

$$f(F_k(X_1, \ldots, X_n)) \equiv F_k(g(X_1), \ldots, g(X_n)) (\bmod \deg(k+1))$$

And the existence and uniqueness of $F(X_1, \ldots, X_n)$ follows from this.

For $k = 1$, we take $F_1(X_1, \ldots, X_n) = L(X_1, \ldots, X_n)$.

Now assume that the result holds for k. Write $f(F_k(X_1, \ldots, X_n)) \equiv F_k(g(X_1), \ldots, g(X_n)) + E_{k+1} (\bmod \deg(k+2))$. According to hypothesis assumption, E_{k+1} is homogeneous of degree $k+1$.

On the other hand, since the uniqueness of $F_k(X_1, \ldots, X_n)$, then $F_{k+1}(X_1, \ldots, X_n)$ should be of forms $F_{k+1}(X_1, \ldots, X_n) = F_k(X_1, \ldots, X_n) + \phi_{k+1}(X_1, \ldots, X_n)$, where $\phi_{k+1}(X_1, \ldots, X_n)$ is homogeneous of degree $k+1$, due to the definition of $\mathcal{F}_\pi$. Then we have

$$f(F_{k+1}(X_1, \ldots, X_n)) \equiv F_k(g(X_1), \ldots, g(X_n)) + \pi \phi_{k+1}(X_1, \ldots, X_n) (\bmod \deg(k+2))$$

$$F_{k+1}(g(X_1), \ldots, g(X_n)) \equiv F_k(g(X_1), \ldots, g(X_n)) + \pi^{k+1} \phi_{k+1}(X_1, \ldots, X_n) (\bmod \deg(k+2))$$

The last congruence is due to ϕ_{k+1} is homogeneous of degree $k+1$, i.e. $\phi_{k+1}(g(X_1), \ldots, g(X_n)) = \pi^{k+1} \phi_{k+1}(X_1, \ldots, X_n)$. Therefore, we conclude that

$$E_{k+1} + (\pi - \pi^{k+1}) \phi_{k+1}(X_1, \ldots, X_n) = 0 \tag{2.1.2.3}$$

Then $\phi_{k+1}(X_1, \ldots, X_n) = -E_{k+1}/(\pi - \pi^{k+1})$. □

Definition 2.1.3. For any $f \in \mathcal{F}_\pi$, we let $F_f(X,Y)$ be the unique solution of

$$F_f(X,Y) \equiv X + Y (\text{mod } \deg 2) \tag{2.1.3.1}$$
$$f(F_f(X,Y)) = F_f(f(X), f(Y)) \tag{2.1.3.2}$$

Theorem 2.1.4 (Lubin-Tate Formal Group Laws)**.** *For power series $f \in \mathcal{F}_\pi$, we have*

$$F_f(X,Y) = F_f(Y,X) \tag{2.1.4.1}$$
$$F_f(F_f(X,Y),Z) = F_f(X, F_f(Y,Z)) \tag{2.1.4.2}$$

As a result, $F_f(X,Y)$ is a commutative formal group law over $\mathcal{O}_K$.

Proof. For equation 2.1.4.1, let $G(X,Y) = F(Y,X)$, then $G(X,Y) \equiv X + Y (\text{mod} \deg 2)$ and

$$f(G(X,Y)) = f(F(Y,X)) = F(f(Y), f(X)) = G(f(X), f(Y))$$

Therefore, by the uniqueness in Lemma 2.1.2, $F(X,Y) = G(X,Y) = F(Y,X)$.

For equation 2.1.4.2, let $G_1(X,Y,Z) = F_f(F_f(X,Y),Z)$ and $G_2(X,Y,Z) = F_f(X, F_f(Y,Z))$, we could check that $G_i(X,Y,Z) \equiv X + Y + Z (\text{mod} \deg 2)$ and

$$f(G_i(X,Y,Z)) = G_i(f(X), f(Y), f(Z))$$

Then by the uniqueness in Lemma 2.1.2, we conclude the desire equation. □

Definition 2.1.5. For each $a \in \mathcal{O}_K$ and $f, g \in \mathcal{F}_\pi$, let $[a]_{f,g}(X)$ be the unique solution of

$$[a]_{f,g}(X) \equiv aX (\text{mod} \deg 2) \tag{2.1.5.1}$$
$$f([a]_{f,g}(X)) = [a]_{f,g}(g(X)) \tag{2.1.5.2}$$

In particular, denote $[a]_f$ instead of $[a]_{f,f}$ to simplify the notation.

Theorem 2.1.6. *For each $a \in \mathcal{O}_K$ and $f, g \in \mathcal{F}_\pi$, the power series $[a]_{f,g}(X)$ is a formal group homomorphism from $F_g(X,Y)$ to $F_f(X,Y)$, i.e. $[a]_{f,g}F_g(X,Y) = F_f([a]_{f,g}(X), [a]_{f,g}(Y))$.*

Proof. For simplicity, we denote $[a]_{f,g}$ as h. We have $h(F_g(X,Y)) \equiv aX + aY (\text{mod} \deg 2)$, $F_f(h(X), h(Y)) \equiv aX + aY (\text{mod} \deg 2)$ and

$$h \circ F_g(g(X), g(Y)) = h \circ g(F_g(X,Y)) = f \circ h(F_f(X,Y))$$
$$F_f(h(g(X), h(g(Y)))) = F_f(f(h(X)), f(h(Y))) = f \circ F_f(h(X), h(Y))$$

Then the result follows from the uniqueness in the Lemma 2.1.2. □

Theorem 2.1.7. *Let $a, b \in \mathcal{O}_K$, $f, g, h \in \mathcal{F}_\pi$, then $[a]_{f,g}([b]_{g,h}(X)) = [ab]_{f,h}(X)$.*

Proof. We have $[a]_{f,g}([b]_{g,h}(X)) \equiv abX (\text{mod} \deg 2)$ and

$$f \circ ([a]_{f,g} \circ [b]_{g,h}) = [a]_{f,g} \circ (g \circ [b]_{g,h}) = [a]_{f,g} \circ [b]_{g,h} \circ h$$

Then we have $[ab]_{f,h}$ and $[a]_{f,g} \circ [b]_{g,h}$ are solutions satisfying these conditions. By the uniqueness of Lemma 2.1.2, they coincide. □

Theorem 2.1.8. *Let $a, b \in \mathcal{O}_K$, $f, g \in \mathcal{F}_\pi$, then $F_f([a]_{f,g}(X), [b]_{f,g}(X)) = [a+b]_{f,g}(X)$.*

Proof. Notice that $F_f([a]_{f,g}(X), [b]_{f,g}(X)) \equiv (a+b)X (\operatorname{mod} \deg 2)$ and

$$f(F_f([a]_{f,g}(X), [b]_{f,g}(X))) = F_f(f \circ [a]_{f,g}(X), f \circ [b]_{f,g}(X)) = F_f([a]_{f,g} \circ f(X), [b]_{f,g} \circ f(X))$$

Therefore, by the uniqueness in Lemma 2.1.2, we have $F_f([a]_{f,g}(X), [b]_{f,g}(X)) = [a+b]_{f,g}(X)$. □

Theorem 2.1.9. *Let $f, g \in \mathcal{F}_\pi$, then F_f is isomorphic to F_g, and there is a unique strong isomorphism from F_f to F_g. This means that the Lubin-Tate formal group F_f is independent of $f \in F_\pi$, depends only on π.*

Proof. Take a unit $a \in \mathcal{O}_K$, then $[a]_{f,g} \circ [a^{-1}]_{g,f} = [1]_f$ and $[a^{-1}]_{g,f} \circ [a]_{f,g} = [1]_g$. And the unique strong isomorphism is $[1]_{f,g}$. □

Remark 2.1.10. Let L be an algebraic extension of K, let $\mathfrak{m}_L$ be the maximal ideal in the ring of integers of L. Given elements $x_1, \ldots, x_n \in \mathfrak{m}_L$ and a formal series $G(X_1, \ldots, X_n) \in \mathcal{O}_K[\![X_1, \ldots, X_n]\!]$, the series $G(x_1, \ldots, x_n)$ will converge an element in $\mathfrak{m}_L$ if the constant term of $G(X_1, \ldots, X_n)$ is zero. By above theorem, for each power series $f \in \mathcal{F}_\pi$ this determines an $\mathcal{O}_K$-module structure on the set $\mathfrak{m}_L$. To be precisely, addition and scalar multiplication are defined by

$$x + y = F_f(x, y) \tag{2.1.10.1}$$

$$ax = [a]_f(x) \tag{2.1.10.2}$$

2.2 Application of Lubin-Tate formal groups

Fix an algebraic closure of $\overline{K}$ of K and K^{sep} be a separable closure. Given a $f \in \mathcal{F}_\pi$, for each integer $n \geq 1$, let $f^n(X)$ be n-th iterate of $f(X)$. Let $\Lambda_{f,n}$ be the set of all roots of $f^n(X)$ in $\overline{K}$ and let $K(\Lambda_{f,n})$ be the field generated over K by these elements. Moreover, we have the fact that for any $f, g \in \mathcal{F}_\pi$, $\lambda \in \Lambda_{f,m}$ if and only if $[1]_{g,f}(\lambda) \in \Lambda_{g,m}$. This means that field $K(\Lambda_{f,n})$ is independent of f and it depends only on uniformizer π. Therefore, we denote $K(\Lambda_{f,n})$ as $L_{\pi,n}$. The same argument holds for $K(\Lambda_f)$. Additionally, let $\Lambda_f = \bigcup_{n=1}^{\infty} \Lambda_{f,n}$, $L_\pi = K(\Lambda_f)$, $G_{\pi,n} = \operatorname{Gal}(L_{\pi,n}/K)$, $G_\pi = \varprojlim G_{f,n}$.

According to the remark 2.1.10, F_f defines a $\mathcal{O}_K$ structure on $\mathfrak{m}_{K^{sep}}$ and we denote this $\mathcal{O}_K$-module as $M_f(K^{sep})$. Then $\Lambda_{f,n}$ is those elements in $\lambda \in \mathfrak{m}_{K^{sep}}$ such that $\pi^n \lambda = 0$.

Theorem 2.2.1. *The $\mathcal{O}_K$-module $\Lambda_{f,n}$ is isomorphic to $\mathcal{O}_K/\pi^n\mathcal{O}_K$ and $\mathcal{O}_K$-module Λ_f is isomorphic to $K/\mathcal{O}_K$.*

In order to prove this theorem, we need a result about divisible module over PID.

Theorem 2.2.2 (Structure theorem of divisible module over PID)**.** *Let R be a principal ideal domain and K be its field of fractions. Then every divisible R-module is isomorphic to a direct sum of K and copies of the injective hulls of simple module $R/(p)$ varying prime elements $p \in R$.*

Let M be a divisible R-module, then M can be written as follows:

$$M \cong (\bigoplus_{i \in I} K) \oplus (\bigoplus_{p \text{ prime}} \bigoplus_{j \in J_p} E(R/(p)))$$

- *I is an index set measuring how many copies of torison free part in M are needed.*

- *p varies over all prime elements in R.*
- *J_p is an index set measuring how many copies of p-primary torison part $E(R/(p))$ are needed for each p.*
- *$E(R/(p))$ is the injective hull of $R/(p)$. Over PID, this is isomorphic to $K/R_{(p)}$.*

Proof. The proof of structure of divisible module can be found in chapter 3 in [1]. □

Proof of theorem 2.2.1. The first claim follows by the fact that $\Lambda_{f,n}$ is the π^n-torsion of $\mathcal{O}_K$-module $M_f(K^{sep})$.

According to theorem 2.1.9, we are free to choose $f \in \mathcal{F}_\pi$ as we please. We take $f(X) = \pi X + X^q$. Since $f'(X) = qX^{q-1} + \pi \neq 0$, then $\Lambda_{f,1}$ consists of q distinct roots of the $X^q + \pi X = 0$. For any given $\alpha \in \mathfrak{m}_{K^{sep}}$, since $\pi X + X^q = \alpha$ is separable, then $\Lambda_{f,1} \subset \mathfrak{m}_{K^{sep}}$. This means that for any given $\alpha \in \mathfrak{m}_K^{sep}$, there exist a β such that $\pi \cdot \beta = \alpha$ in $M_f(K^{sep})$. It follows that for any given $r \in \mathcal{O}_K$, $rM_f(K^{sep}) = M_f(K^{sep})$ which means that $M_f(K^{sep})$ is divisible $\mathcal{O}_K$-module. Then Λ_f is a divisible torsion $\mathcal{O}_K$-module. Since $\mathcal{O}_K$ is a PID, then we know that Λ_f is isomorphic to $K/\mathcal{O}_K$. Indeed, Λ_f is torsion module then there is no copies of K. Since π is the only prime element in $\mathcal{O}_K$ and $(\mathcal{O}_K)_{(\pi)} \cong \mathcal{O}_K$, then Λ_f is a copies of $K/\mathcal{O}_K$. Moreover, $\Lambda_{f,1}$ is a one dimensional vector space over residue field k (since $\Lambda_{f,1}$ has the same cardinality of residue field k), then $\Lambda_f \cong K/\mathcal{O}_K$. Last claim follows by the fact that $|J| = \dim_k M_f(K^{sep})[\pi] = \dim_k \Lambda_{f,1} = 1$, where J is the index set of copies of $K/\mathcal{O}_K$. □

There is a natural homomorphism $\mathrm{Gal}(K^{sep}/K) \to \mathrm{End}(\Lambda_f)$, this is due to the properties by the action of element in a Galois group on roots of a polynomial. Moreover, since $\Lambda_f \cong K/\mathcal{O}_K$ and $\mathrm{End}(K/\mathcal{O}_K) \cong \mathcal{O}_K$, then it induces a homomorphism $\mathrm{Gal}(K^{sep}/K) \to \mathrm{Aut}(K/\mathcal{O}_K) \cong U_K$.

Theorem 2.2.3. *The natural homomorphism* $\mathrm{Gal}(K^{sep}/K) \to \mathrm{Aut}(\Lambda_f)$ *is an isomorphism.*

Proof. This map is injection due to our definition, then it remain to prove it is a surjection. We have an injection $\mathrm{Gal}(L_{\pi,n}/K) \to U_K/U_K^n$, where $U_K = 1 + \pi^n \mathcal{O}_K$. Let $\alpha \in \Lambda_{f,n}$ be a primitive element, namely those elements such that $[\pi^{n-1}]_f \alpha \neq 0$ or $\pi^{n-1} \cdot \alpha \neq 0$ in $M_f(K^{sep})$. To be precise, α is one of the roots of polynomial $f^{(n)}(X)/f^{(n-1)}(X)$, where $f^{(n)}(X)$ is n-th iterate of $f(X)$. Since $f(X)/X = X^{q-1} + \pi$, then

$$\frac{f^{(n)}(X)}{f^{(n-1)}(X)} = f^{(n-1)}(X)^{q-1} + \pi$$

By the Eisenstein criterion, this is an irreducible polynomial with degree $q^n - q^{n-1}$. Thus, according to the Galois theory, the order of $\mathrm{Gal}(L_{\pi,n}/K)$ is at least $(q-1)q^{n-1}$. On the other hand, U_K/U_K^m has exactly $(q-1)q^{n-1}$ elements. Therefore, $\mathrm{Gal}(L_{\pi,n}/K) \cong U_K/U_K^n$. Eventually, we have that

$$\mathrm{Gal}(K^{sep}/K) \cong \varprojlim \mathrm{Gal}(L_{\pi,n}/K) \cong \varprojlim U_K/U_K^n \cong U_K.$$

□

Corollary 2.2.4. *The element π is a norm from $L_{\pi,n}$.*

Proof. Let $\alpha \in \Lambda_{f,n}$ be a primitive element, then $f^{(n)}(X)/f^{(n-1)}(X)$ is a monic polynomial with constant coefficient π, then the norm $-\alpha$ is π. □

Let K^{un} be the maximal unramified extension of K. Since L_π is totally ramified extension of K, then L_π and K^{un} is linearly disjoint. As a result, the Galois group $\mathrm{Gal}(L_\pi K^{un}/K)$ is a product of $\mathrm{Gal}(L_\pi/K)$ and $\mathrm{Gal}(K^{un}/K)$. Given a uniformizer element $\pi \in \mathcal{O}_K$, we define a group homomorphism $r_\pi : K^\times \to \mathrm{Gal}(L_\pi K^{un}/K)$ such that

- $r_\pi(\pi)$ acts as identity on L_π and acts as Frobenius automorphism σ on K^{un}.
- Given a $u \in U_K$, we define $r_\pi(u)$ acts as $[u^{-1}]_f$ on L_π and acts as identity on K^{un}.

Next we will prove that the field L_π and the homomorphism r_π are independent of the uniformizer π and it is a maximal abelian extension of K. Moreover, r_π coincides with the local reciprocity map in local class field theory, i.e. for all $a \in \mathcal{O}_K$, $r_\pi(a) = (a, L_\pi K^{un}/K^\times)$.

3 Lean formalization of Formal Group Laws

In this chapter we will discuss our formalization of formal group laws.

3.1 Definitions

We choose to formalize definition of formal group laws in Lean using the definition in 1.1.1, since this is equivalent to definitions in other references and more easier to check in practice.

The following is our definition of formal group laws in Lean:

```
@[ext]
structure FormalGroup (R : Type*) [CommRing R] where
  toFun : MvPowerSeries (Fin 2) R
    /- Constant coefficient of `F` is zero. -/
    zero_constantCoeff : constantCoeff toFun = 0
    /- Coefficient of `X` and `Y` is 1. -/
    lin_coeff_X : coeff (single 0 1) toFun = 1
    lin_coeff_Y : coeff (single 1 1) toFun = 1
    /- Associativity condition: `F (F (X, Y), Z) = F (X, F (Y, Z))`.-/
    assoc : subst ![subst ![Y₀, Y₁] toFun, Y₂] toFun = subst ![Y₀, subst ![Y₁, Y₂]
    toFun] toFun (S := R)
```

In Mathlib, `MvPowerSeries` σ `R` is the ring of multivariate power series with intermediates indexed by σ. And `Fin n` is a type and there are n different elements in this type. To be specific, an element in `Fin n` is a `i : ℕ` with a proof `i < n`. And `MvPowerSeries.X (i :` σ `)` is the intermediate indexed by `i` and in name space of `MvPowerSeries` we could abbreviate `MvPowerSeries.X` to be `X`. Therefore, `X 0, X 1, X 2` are all three intermediates in `MvPowerSeries (Fin 3) R`. The `Y₀, Y₁, Y₂` is an abbreviation for `X 0, X 1, X 2` in `MvPowerSeries (Fin 3) R`. The expression `![a, b]` in the associativity condition is a map from `Fin 2` to α where `a b :` α. And `subst` is a function used to express the substitution of (multivariate) power series. For instance, $f(g(X_I))$ is expressed as `f.subst g` and $F(f(X_I), g(X_I))$ is expressed as `F.subst ![f, g]`.

For the definition for commutative formal group laws, the only additional condition is commutativity condition.

```
@[ext]
structure CommFormalGroup (R : Type*) [CommRing R] extends FormalGroup R where
/- Commutativity condition: `F(X,Y) = F(Y,X)`.-/
comm : toFun = subst ![X₁, X₀] toFun
```

Here, `X₀, X₁` is an abbreviation for `X 0, X 1` in `MvPowerSeries (Fin 2) R`.

Mathematically, a commutative formal group is a formal group, this statement is called coercions in Lean. An instance about this is as follows:

```
/-- A commutative formal group law is a formal group law.-/
instance : Coe (CommFormalGroup R) (FormalGroup R) where
coe := CommFormalGroup.toFormalGroup
```

Given a formal group law, it defines a group structure on `MvPowerSeries σ R`. To be precise, let `(F : FormalGroup R) (f₀ f₁ : MvPowerSeries σ R)`, the addition of `f₀ f₁` is defined as `subst ![f₀, f₁] F.toFun`. In Lean, a simpler notation related to this can written as,

```
abbrev add (F : FormalGroup R) (f g : MvPowerSeries σ R) : MvPowerSeries σ R :=
    subst ![f, g] F.toFun
```

In the environment of `namespace FormalGroup`, `+[F]` is the addition with respect to `F : FormalGroup R`. To be specific, given two multivariate power series $f, g \in R[\![X_I]\!]$ where I is index set, and the constant coefficient of f, g are zero, `f +[R] g` means $F(f, g)$.

With this notation, we can rewrite the associativity condition and the commutativity condition in the `namespace FormalGroup` as follows:

```
/-- The addition under the sense of formal group `F` is associative. -/
theorem add_assoc {F : FormalGroup R} {Z₀ Z₁ Z₂ : MvPowerSeries σ R} (hZ₀ :
    constantCoeff Z₀ = 0) (hZ₁ : constantCoeff Z₁ = 0) (hZ₂ : constantCoeff Z₂ = 0):
Z₀ +[F] Z₁ +[F] Z₂ = Z₀ +[F] (Z₁ +[F] Z₂) := ...

theorem add_comm {F : FormalGroup R} (hF : F.comm) {Z₀ Z₁ : MvPowerSeries σ R}
(hZ₀ : constantCoeff Z₀ = 0) (hZ₁ : constantCoeff Z₁ = 0):
  Z₀ +[F] Z₁ = Z₁ +[F] Z₀ := ...
```

To make sense that formal group law F defines a formal addition, we should verify that there is a unit in the addition group, which means $F(X, 0) = X$ and $F(0, Y) = Y$. Furthermore, we should prove there exists a unique (addition) inverse element.

The existence of additional unit in F sense is formalized as the following two theorems.

```
theorem zero_left : subst ![0, PowerSeries.X] F.toFun = PowerSeries.X (R := R) := ...

theorem zero_right : subst ![PowerSeries.X, 0] F.toFun = PowerSeries.X (R := R) := ...
```

Using the same notation of formal group addition, I generalized the theorems about to the following two theorems.

```
theorem zero_add (h : constantCoeff f = 0) : 0 +[F] f = f :=

theorem add_zero (h : constantCoeff f = 0) : f +[F] 0 = f :=
```

Here, the name `zero_add` means that zero adds some a multivariate power series `f` equal to `f` itself , which is a standard naming convention in Mathlib. In order to distinguish this theorem to other theorem with the same name, we put this theorem in the `namespace FormalGroup`

3.1.1 Examples of formal groups

The basic examples of formal group laws are additive formal group law $\mathbf{G}_a$ and multiplicative formal group law $\mathbf{G}_m$. We formalized them via our definition of `FormalGroup R`

```
def Gₐ : CommFormalGroup R where
  toFun := X₀ + X₁
  ...

def Gₘ : CommFormalGroup R where
  toFun := X₀ + X₁ + X₀ * X₁
  ...
```

3.1.2 Function between formal groups

According to 1.1.7, given two commutative ring R, R' and a ring homomorphism $\phi : R \to R'$, let a formal group law F over R, then $\phi_* F$ is a formal group law over R'. The construction of the push forward of the coefficient ring in Lean:

```
def map {R : Type*} [CommRing R] {R' : Type*} [CommRing R']
    (f : R →+* R') (F : FormalGroup R): FormalGroup R' where
  toFun := F.toFun.map f
  ...
```

Let R be a commutative ring and T be a subring of R. Given a (multivariate) power series f over R with the assumption that all coefficients of f in the subring of T, then f can be considered as a (multivariate) power series over T. We define this function as `MvPowerSeries.toSubring` which follow the APIs in polynomial as inspiration.

```
variable [Ring R] (p : MvPowerSeries σ R) (T : Subring R) (hp : ∀ n, p.coeff n ∈ T)

def MvPowerSeries.toSubring : MvPowerSeries σ T := fun n => ⟨p.coeff n, hp n⟩
```

According to the definition, the coefficients of them should be the same after coercion. And the `@[simp]` tag in the following theorem means that this theorem could be automatically applied if we use tactic `simp` to simplify the expression.

```
@[simp]
theorem coeff_toSubring {n : σ →₀ ℕ} : (p.toSubring T hp).coeff n = p.coeff n := rfl
```

Given a formal group law F over ring R with the assumption that all coefficients of F are in the subring T, then F is a formal group law over T.

```
def FormalGroup.toSubring (F : FormalGroup R) (hF : ∀ n, F.toFun n ∈ T) :
    FormalGroup T where
  toFun := F.toFun.toSubring _ hF
  ...
```

Similarly, we could also define `CommFormalGroup.toSubring`. Since we have a coercion from `CommFormalGroup R` to `FormalGroup R`, we only need to check commutativity condition.

```
lemma CommFormalGroup.toSubring_comm (F : CommFormalGroup R) (hF : ∀ n, F.toFun n ∈
    T) :
    F.toFun.toSubring _ hF = (F.toFun.toSubring _ hF).subst ![X₁, X₀] := ...

def CommFormalGroup.toSubring (F : CommFormalGroup R) (hF : ∀ n, F.toFun n ∈ T) :
    CommFormalGroup T where
  toFun := F.toFun.toSubring _ hF
  zero_constantCoeff := ((F : FormalGroup R).toSubring _ hF).zero_constantCoeff
  lin_coeff_X := ((F : FormalGroup R).toSubring _ hF).lin_coeff_X
  lin_coeff_Y := ((F : FormalGroup R).toSubring _ hF).lin_coeff_Y
  assoc := ((F : FormalGroup R).toSubring _ hF).assoc
  comm := F.toSubring_comm _ hF
```

3.1.3 Homomorphism and Isomorphism

Firstly, we need the API for `PowerSeries.toMvPowerSeries` which is inspired by a similar API in polynomial.

```
noncomputable
def Polynomial.toMvPolynomial (i : σ) : R[X] →ₐ[R] MvPolynomial σ R :=
  aeval (MvPolynomial.X i)
```

Given a power series $p(X) \in R[\![X]\!]$ and an index $i \in I$ where I is index set, this is a function mapping $p(X)$ to $p(X_i) \in R[\![X_I]\!]$.

```
def PowerSeries.toMvPowerSeries (f : PowerSeries R) (i : σ) : MvPowerSeries σ R :=
  f.subst (MvPowerSeries.X i)
```

For the definition of homomorphism 1.1.5 between two formal group laws, it is defined as `structure` in Lean:

```
@[ext]
structure FormalGroupHom  (G₁ G₂ : FormalGroup R) where
  toFun : PowerSeries R
  zero_constantCoeff : toFun.constantCoeff = 0
  hom : toFun.subst (G₁.toFun) = G₂.toFun.subst (R := R) toFun.toMvPowerSeries
```

Since two formal group homomorphisms are the same if and only if their underlying functions are the same. Therefore, we add a tag `@[ext]` to the structure. When `@[ext]` is applied to a structure, it generates an `.ext` theorem and applies the `@[ext]` attribute to it. The result is an `.ext` and an `.ext_iff` theorem with the `.ext` theorem registered for the `ext` tactic.

The isomorphism of two formal groups is defined as two formal group homomorphisms which are inverse to each other.

```
@[ext]
structure FormalGroupIso (G₁ G₂ : FormalGroup R) where
  toHom : FormalGroupHom G₁ G₂
  invHom : FormalGroupHom G₂ G₁
  left_inv : (PowerSeries.subst toHom.toFun) ∘ (PowerSeries.subst invHom.toFun) = id
  right_inv : (PowerSeries.subst invHom.toFun) ∘ (PowerSeries.subst toHom.toFun) = id
```

Here we also add a tag `@[ext]` for the `structure` `FormalGroupIso` and we will automatically have a theorem

```
variable {G₁ G₂ : FormalGroup R} {x y : FormalGroupIso G₁ G₂}

theorem FormalGroupIso.ext_iff :
    x = y ↔ x.toHom = y.toHom ∧ x.invHom = y.invHom := ...
```

To show that two formal group isomorphisms are the same, it is enough to show that their `toHom` is the same. Indeed, if we have `x.toHom = y.toHom`, then `x.invHom = y.invHom` are the same since `invHom` is the substitution inversion of `toHom` and the substitution inverse is unique. And the following theorem is more convenient to use in practice.

```
theorem FormalGroupIso.ext_iff' : x = y ↔ x.toHom = y.toHom := ...
```

3.2 Functional Equation Lemma

In this section, we will discuss the formalization of functional equation lemma. To begin with, we formalize the datum involved in the functional equation lemma as follow.

```
variable {K : Type*} [CommRing K] {R : Subring K} {I : Ideal R}
  {p t q : ℕ} [hp : Fact (Nat.Prime p)] (ht : t ≠ 0) (hq : q = p ^ t)
  (σ : K →+* K) (s : ℕ → K) {g : PowerSeries R}
```

Let K be a commutative ring, R be a subring of K, I be a ideal of R. Let p, q, t be natural number, where p is a prime number and q can be written as $q = p^t$ with a nonzero t. Let σ be a ring endomorphism of K and s is a series of elements in K.

According to the equation 1.2.2.1, we can inductively define a function `RecurFunAux` using equation 1.2.2.1 and obtain a power series using the API `PowerSeries.mk`, then prove it satisfy the functional equation 1.2.1.1, namely

$$f_g(X) = g(X) + \sum_{i=1}^{\infty} s_i \sigma_*^i f_g(X^{q^i}).$$

```
def RecurFunAux (hg : g.constantCoeff = 0) : ℕ → K
  | 0 => 0
  | k + 1 =>
    g.coeff (k + 1) + ∑ j ∈ (Icc 1 (multiplicity q (k + 1))).attach,
      have : ((k + 1) / (q ^ (j : ℕ))) < k + 1 := ...
      (s j) * σ^[j] (RecurFunAux hg ((k + 1) / (q ^ (j : ℕ))))

def RecurFun := PowerSeries.mk (RecurFunAux ht hq σ s hg)
```

In order to inductively define `RecurFunAux`, we prove that for all `j ∈ Icc 1 (multiplicity q (k + 1))`, `((k + 1) / (q ^ (j : ℕ))) < k + 1` in the motive of `k + 1`. Therefore, we need to use `Finset.attach` to enable us to use the property of `j ∈ Icc 1 (multiplicity q (k + 1))`. To be precise, let `s` be a finite set, `Finset.attach s` takes the elements of `s` and forms a new set of elements of the subtype `{x // x ∈ s}`. With the property `j ∈ Icc 1 (multiplicity q (k + 1))`, we can prove that `((k + 1) / (q ^ (j : ℕ))) < k + 1`. Otherwise, `(RecurFunAux hg ((k + 1) / (q ^ (j : ℕ))))` is not valid.

To state the formalization of functional equation, we need APIs related to `expand`, which is a algebraic homomorphism that expand a (multivariate) power series by a nonzero factor p. We will discuss these APIs first.

```
variable {σ τ R S : Type*}  [CommRing R] [CommRing S] (p : ℕ) (hp : p ≠ 0)

/-- Expand the power series by a factor of p, so `∑ aₙ x^n` becomes `∑ aₙ x^(np)`. -/
noncomputable def expand : MvPowerSeries σ R →ₐ[R] MvPowerSeries σ R :=
  substAlgHom (HasSubst.X_pow hp)
```

We formalize functional equation 1.2.1.1 as follow.

```
theorem FunEq_of_RecurFun [TopologicalSpace K] [T2Space K] (hs₀ : s 0 = 0) :
    let f := RecurFun ht hq σ s hg
    f = g.map R.subtype + ∑' (i : ℕ), s i · (f.expand (q ^ i) (q_pow_neZero hq)).map
    (σ^i) := ...
```

The additional instance `[TopologicalSpace K]` provides a topological structure on ring of power series over K and `[T2Space K]` ensure that the infinite sum in the equation has unique limit, whenever it converges. Let R be a subring of K, then `R.subtype` is a canonical ring homomorphism from R to K. `PowerSeries.expand p hp f` expands a power series `f` by a factor of nonzero p.

The theorem `coeff_one_RecurFun` is the 1-th coefficient of g equals to the 1-th coefficient of $f_g(X)$. With the assumption that `(hg_unit : IsUnit (g.coeff 1))`, then `IsUnit ((RecurFun ht hq σ s hg).coeff 1)`. Therefore, there is a substitution inverse of $f_g(X)$ which denote as $f_g^{-1}(X)$.

```
lemma coeff_one_RecurFun: (RecurFun ht hq σ s hg).coeff 1 = g.coeff 1 := ...

lemma coeff_RecurFun_unit (hg_unit : IsUnit (g.coeff 1)) :
    IsUnit ((RecurFun ht hq σ s hg).coeff 1) := ...

def inv_RecurFun := PowerSeries.subst_inv _ (coeff_RecurFun_unit ht hq σ s hg hg_unit)
  (constantCoeff_RecurFun ..)

def inv_add_RecurFun :=
    (inv_RecurFun ht hq σ s hg hg_unit).subst
      ((RecurFun ht hq σ s hg).toMvPowerSeries (0 : Fin 2) +
        (RecurFun ht hq σ s hg).toMvPowerSeries 1)
```

The API `inv_add_RecurFun` is the multivariate power series $F_g(X,Y) = f_g^{-1}(f_g(X) + f_g(Y))$. And `(RecurFun ht hq σ s hg).toMvPowerSeries (0 : Fin 2)` represents the multivariate power series $f_g(X)$, where X is the first indeterminate variable in two variate power series. Similarly, `(RecurFun ht hq σ s hg).toMvPowerSeries 1` represents $f_g(Y)$.

Recall that the first part of functional equation lemma 4 is that $F_g(X,Y)$ has its coefficient in R. Moreover, this is a commutative formal group law due to the definition of $F_g(X,Y)$. The first part of functional equation lemma is formalized as follow.

```
theorem coeff_inv_add_mem_Subring [UniformSpace K] [T2Space K] [DiscreteUniformity K]
    (hs₀ : s 0 = 0) : ∀ n, (inv_add_RecurFun ht hq σ s hg hg_unit).coeff n ∈ R := ...
```

Next, we first prove that $F_g(X,Y)$ is a formal group law over K. The associativity condition follows that

$$F_g(F_g(X,Y),Z) = f_g^{-1}(f_g(X) + f_g(Y) + f_g(Z)) = F_g(X, F_g(Y,Z))$$

Then we use the API `FormalGroup.toSubring` and the theorem `coeff_inv_add_mem_Subring` to get a formal group law `CommFormalGroup.InvAdd_RecurFun` over R.

```
def CommFormalGroup.InvAdd_RecurFun_Aux : CommFormalGroup K where
  toFun := inv_add_RecurFun ht hq σ s hg hg_unit
  ...

def CommFormalGroup.InvAdd_RecurFun [UniformSpace K] [T2Space K]
    [DiscreteUniformity K] (hs₀ : s 0 = 0) : CommFormalGroup R :=
  (CommFormalGroup.InvAdd_RecurFun_Aux ht hq σ s hg hg_unit).toSubring _
    (coeff_inv_add_mem_Subring ...)
```

Recall that the second part of the functional equaion lemma 4 is that for any power series g, g' over R with the assumption that the constant coefficients of g, g' is zero, then $f_{g'}^{-1}(f_g(X))$. Then the second part of the functional equation lemma is formalized as follow. In the lemma below, `f_g'` is the power series $f_{g'}(X)$ and the `G` is $f_{g'}^{-1}(f_g(X))$.

```
lemma coeff_inv_RecurFun_g'_mem_Subring [UniformSpace K] [T2Space K]
    [DiscreteUniformity K] (hs0 : s 0 = 0) :
    let f_g' := (RecurFun ht hq σ s hg')
    let G := (inv_RecurFun ht hq σ s hg hg_unit).subst f_g'
    ∀ n, PowerSeries.coeff n G ∈ R
```

The third part of the functional equation lemma is that there is a power series $\hat{h}$ over R, such that $f_g(h(X)) = f_{\hat{h}}(X)$. According to the functional equation 1.2.1.1, it is equivalent to show that the power series $f_g(h(X)) - \sum_{n=1}^{\infty} s_n \cdot \sigma^n f_g(h(X^{q^n}))$ is a power series over R. And this is exactly the desired power series $\hat{h}$.

```
lemma coeff_inv_RecurFun_g'_mem_Subring' [UniformSpace K] [T2Space K]
    [DiscreteUniformity K] (hs₀ : s 0 = 0) {h : PowerSeries R}
    (h₀ : h.constantCoeff = 0):
    let f := (RecurFun ht hq σ s hg)
    let f₁ : PowerSeries K := f.subst (h.map R.subtype)
    ∀ n, (f₁ - ∑' i, (s i) · (f₁.expand (q ^ i) (q_pow_neZero hq)).map (σ ^ i)).coeff
    n ∈ R :=

theorem congr_equiv [UniformSpace K] [T2Space K] [DiscreteUniformity K]
    (hs₀ : s 0 = 0) {α : PowerSeries R} {β : PowerSeries K}
    (hα : α.constantCoeff = 0) (hβ : β.constantCoeff = 0)
    {r : ℕ} (hr : 1 ≤ r) :
    let f := RecurFun ht hq σ s hg
    (∀ n, α.coeff n - β.coeff n ∈ R.subtype '' ↑(I ^ r)) ↔
      (∀ n, PowerSeries.coeff n (f.subst (α.map R.subtype)) -
        PowerSeries.coeff n (f.subst β) ∈ R.subtype '' ↑(I ^ r)) :=
```

I would explain some details in part four of the functional equation lemma briefly. To begin with, `α.coeff n - β.coeff n` is a element in K, since α is a power series over R and β is a power series over K. Then we need to formalize it as a element in a set of K rather than in ideal of R. Secondly, `R.subtype '' ↑(I ^ r)` is a set of elements in K which is image of I^r under the natural algebraic map $R \to K$. Moreover, the symbol ↑ is necessary since there is a coercion from `Ideal R` to `Set R`.

3.3 Universal formal group laws

3.4 Commutative Lemma

3.5 Lubin-Tate Theory

Bibliography

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